Fiscal Year:	FY 2012	**Task Last Updated:**	FY 06/03/2012
PI Name:	Hammond, Timothy M.D.		
Project Title:	Genotypic and Phenotypic Changes in Yeast Related to Selective Growth Pressures Unique To Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	None		
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Comments:	Alternate email: Timothy.Hammond@va.gov NOTE: PI previously at Tulane University until 2007 (Ed., 09/2010) Joint appointments Professor, Department of Internal Medicine, Nephrology Division, Duke University School of Medicine, and Staff Physician, Durham VA Health Care System.		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	08/01/2010	**End Date:**	07/31/2013
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
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Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-135 ; ISS NOTE: Extended to 7/31/2013 per A. Chu/ARC and NSSC information (Ed., 6/03/2012)		
Key Personnel Changes/Previous PI:	June 2012: Add Dr. Ricardo Saban, a subject matter expert in gene array data analysis, as a consultant. Add Dr. Holly Birdsall as a data warden (co-investigator) so all analysis can be performed blinded.		
COI Name (Institution):	Birdsall, Holly Hyde (Department of Veterans Affairs)		
Grant/Contract No.:	NNX10AP01G		
Performance Goal No.:			
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Task Description:	This application aims to investigate how cells adapt to the unique aspects of the space environment, using the model eukaryotic organism, Saccharomyces cerevisiae. Direct quantitative assessment of selective pressures on cell populations through generations has become possible due to the recent availability of the yeast deletion series. This powerful reagent is a mixture of molecularly engineered isogenic yeast strains that differ only in a single gene locus. Each gene has been replaced with an identifying "bar code" and the mixture contains a deletion strain for every gene in the yeast genome. The yeast deletion series is grown under a selective pressure, DNA extracted, and the barcodes amplified by PCR. The resulting product is annealed to a gene microarray chip comprising spots for the complementary sequence of each barcode. In this manner the selective growth of every strain in the yeast deletion series can be assayed in a single tube, enabling a genomic approach to phenotypic analyses. These studies have a strong link and a clear path to flight research on the International Space Station because of the properties of the yeast deletion series: it can be flown in a quiescent state, and activated by simple fluid addition and completed by fixative addition. Importantly, the post-flight analysis is of DNA, which is orders of magnitude easier to fix and preserve than RNA. Given these characteristics, we propose a flight experiment, with a high probability of obtaining genome wide data. This flight experiment directly addresses two broad areas Fundamental Space Biology is attempting to expand: 1) Life at molecular and cellular levels, & 2) Life across generations To mimic some of the elements of microgravity on the ground, in a proof-of-concept experiment we propose to utilize the rotating wall vessel, a form of suspension culture which mimics some elements associated with microgravity, namely random orientation to the gravitational vector, and falling at terminal velocity. Sets of the rotating wall vessel grown under conventional means will control for rotational effects, and spotting the yeast series on agar plates controls for the effects of suspension culture. Our hypothesis is that yeast cells will undergo a selective response to growth conditions that reflect the space environment and that this phenotypic response will be controlled by changes in gene expression of specific target genes. Based on our preliminary data our analysis will examine both specific pathways such as ones utilizing the Stress Responsive Element (STRE), as well as ones that test for novel pathways. The steps we will take to examine this hypothesis are as follows below: 1. Identify those yeast genes regulating growth under the selective pressure of the rotating wall vessel, using the yeast deletions strains 2. Confirm the ability of the specific regulatory motifs to directly activate transcription of a set of reporter genes 3. Identify those trans-acting factors that interact with these regulatory motifs 4. Confirm and extend the results obtained in our ground based studies on the selective growth pressure in a true microgravity environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Space based growth of microorganisms appears to induce a unique set of virulence factors, which may identify gene pathways which can be deleted to generate weakened bacterial forms suitable for vaccine development.
	This project aims to investigate how cells adapt to the unique aspects of the space environment, using the model eukaryotic organism, Saccharomyces cerevisiae. Our hypothesis is that yeast cells will undergo a selective response to growth conditions that reflect the space environment and that this phenotypic response will be controlled by changes in gene expression of specific target genes. The specific aims are: 1. Identify those yeast genes regulating growth under the selective pressure of the rotating wall vessel, using the yeast deletions strains ; 2. Confirm the ability of the specific regulatory motifs to directly activate transcription of a set of reporter genes ; 3 Identify those trans-acting factors that interact with these regulatory motifs ; 4. Confirm and extend the results obtained in our ground based studies on the selective growth pressure in a true microgravity environment. In the grant application we proposed to study yeast fitness profiling using a yeast deletion series and Affymetrix chip array analysis. We also identified an alternative technical approach using colony size as a proxy for fitness using yeast deletions spotted and grown on agar plates. In the time since the application was written, reviewed, and funded, the field has progressed and new methods dramatically increase accuracy and reproducibility, lower error, and lower sample volumes. The grant application proposed comparison of suspension cultures to plate based spotted arrays. Both these techniques have been optimized in the Banting and Best lab headed by Dr. Charlie Boone at the University of Toronto who are kindly collaborating on the project. The suspension cultures can now be analyzed by high throughput sequencing, so that analysis reaches the molecular level, vastly superior to microarray methods due to the accuracy and reproducibilty of the technique. Global quantitative analysis of genetic interactions in spotted plate based assays (see Anastasia Baryshnikova et al. Nature Methods (7) 1017–1024, 2011) is a powerful approach for deciphering the roles of genes and mapping functional relationships among pathways. Using colony size as a proxy for fitness, our collaborators in the lab headed by Charlie Boone developed a method for measuring fitness-based genetic interactions from high-density arrays of yeast double mutants generated by synthetic genetic array (SGA) analysis. They identified several experimental sources of systematic variation and developed normalization strategies to obtain accurate single- and double-mutant fitness measurements. The combined use of spotted plate arrays and suspension cultures allows us to analyze if changes in fitness profiling ion space are primary due to direct cellular efforts of gravity, or secondary to changes in suspension culture physical forces and convection in space. Investigators from the primary science group at the Durham V.A. (Timothy Hammond and Patricia Allen) and the engineering group at Bioserve Space Technologies at University of Colorado – Boulder (Louis Stodieck) have made two visits to the Banting and Best lab headed by Dr. Charlie Boone at the University of Toronto. The first visit was a planning visit to define the experiment. The second visit was to execute the formal Experimental

Task Progress:

Verification Test (EVT).

During the EVT samples were prepared and the flight, ground, and clinostat experimental groups executed. Hardware was loaded in Toronto, Canada, and shipped to Boulder, Colorado in the United States uneventfully. This was facilitated by the fact that the Boone lab routinely and commonly ships yeast to collaborators in the US, and the US maintains a border station in Toronto allowing on site US customs clearance in Canada.

Execution of space flight experiment on STS-135 and post-flight analysis. The space flight experiment successfully completed all experimental procedures and milestones.

Step 1: Transport of yeast to the International Space Station. Yeast was preserved simply by cooling suspension cultures and agar plates to +4oC. Once on ISS, the experiment is initiated by raising the temperature to +30oC.

In preparation for the space flight, yeast was both spotted on plates (plate cultures) and placed in YPD culture medium (suspension culture) then held at +4oC for 8 days, Analysis showed that both the spots and liquid cultures remain quiescent, but grew robustly when the temperature was raised to +30oC. These protocols were formalized in writing for the EVT, and the yeast grew flawlessly as predicted in the EVT. The same parameters were applied for the space flight experiment.

In flight experimental parameters. During experimental mock-ups in the PI laboratories, yeast was cultured in selected hardware.

The culture bags are flight-proven and preliminary ground based mockups have been uncomplicated but interactions between components were tested over the entire time course of the proposed experiments.

The yeast grew robustly in the hardware during the EVT achieving the same optical density (OD)s routinely achieved in well culture in the Boone lab, specifically growth to an OD of 16.

The optimization of the new methods by the lab in Toronto shows that a ratio of 17 microlitres into 700 microlitres is optimal for suspension cultures, and is fully scalable, so this ratio was successfully adopted for the EVT and subsequent flight.

The current protocol used 2 incubators, one on the space shuttle and one on the ISS. The space shuttle incubator remained at +4oC throughout, and the warmed samples transfer to the ISS incubator at +30oC. Mockups demonstrated that the samples both on spotted plates and in liquid culture are stable >2 weeks at +4oC. Sequence analysis of the flight samples is ongoing, but no problems have been encountered to date.

Post-flight payload recovery:

CGBA-2 containing all OPMs and PHabs returned approximately 4 hours after landing. All Opticells were unloaded individually into tubes. Optical density of aliquots showed robust growth of all samples to OD 16. All samples were packed and shipped to University of Toronto on dry ice. For analysis to count deletion copies by sequencing-- DNA was extracted, cDNA generated, and high through put sequencing performed in two cycles of depth.

Photographs were taken of the spotted plates using specialized imaging system which had been sent to KSC, and plates scraped to remove yeast for shipment to University of Toronto at 4oC to confirm photographic findings by high throughput sequencing. Analysis was performed by photograph digital analysis and pathway analysis using automated software. Initial pathway analysis is complete, but being further refined, and a public data website will be active this week.

Genes determining survival advantage in S. cerevisiae changes during static, RWV, and flight plate culture - Deletion series and scanning electron microscopy studies.

On the last space shuttle flight, known as STS-135, our labs flew the yeast deletion series grown in suspension culture and spotted on plates. Scanning electron micrographs show the phenotype of the yeast grown in true microgravity as compared to ground-based microgravity emulation. Analysis of the gene deletions that mediated growth advantage or disadvantage during rotation or flight allows construction of pathway diagrams to summarize the data.

Multiple samples of static ground, clinostat, and flight samples were examined by scanning electron microscopy. This demonstrated differences in shape, surface texture and budding pattern when grown in YPD in Opticells as suspension cultures. Specifically, the static ground samples were egg to barrel shaped single pole budding and undulated surface; the clinostat samples were round to barrel shaped, with multiple bi-polar buds, and massively undulated surface; and, the flight samples were round to egg shaped with multiple buds, large smooth patches, and hugely undulated surface.

Identification of genetic-environmental interactions in the flown and grown samples versus flown-not grown controls. In this analysis the normalized sizes of each of 4 replicates for each single mutant deletion strain was compared to the flown not-grown samples that were grown upon their return. This allowed us to develop a rank order list of 4800 strains which had a significant change in survival during space flight, with magnitude of change determining position on the list. Pathway analysis. For initial analysis the 50 strains with the greatest space-related fitness defects were imported into the meta-analytical tool, GeneMANIA (http://www.genemania.org). GeneMANIA comprises a series of algorithms that searches large, publicly available biological datasets to find related genes. These include protein-protein, protein-DNA and genetic interactions, pathways, re¬actions, gene and protein expression data, protein domains and phenotypic screening profiles. These relationships can be of several types: Co-expression: Two genes are linked if their expression levels are similar across conditions in a gene expression study. Physical Interaction: Two gene products are linked if they were found to interact in a protein-protein interaction study. Genetic interaction: Two genes are functionally associated if the effects of perturbing one gene were found to be modified by perturbations to a second gene. Co-localization: Genes expressed in the same tissue, or proteins found in the same location. Two genes are linked if they are both expressed in the same tissue or if their gene products are both identified in the same cellular location. We then prepared a visual representation of the output from this analysis where each gene is a node in the network, and the links between each node represents one type of interaction between these 50 genes. This preliminary analysis reveals several clusters of functionally interconnected genes, including clusters of genes involved in DNA repair and metabolism, transcription, chromatin remodeling, mitochondrial function, polarity, ribosome assembly, and stress.

The flight sensitive strains features strains deleted for DNA synthesis and repair and mitochondrial function, a result

	reminiscent of treatment of these pools at 1g with DNA damaging agents. The clinostat sensitive deletion strains are not enriched for a particular biological process and are dominated by so-called noisy deletions that appear frequently across many experiments. Flight and RWV sensitive depletion strains: The sequencing counts for each barcode for each deletion mutant (the 4800 homozygous deletions in this experiment) derived from 3 biological replicates were combined and the log2 ratios of sequence counts of the fight samples (at 21 generations of growth) were compared to the same values for the clinostat samples at 21 generations of growth. In this visualization we applied a very strict cut-off of log2=6 such that each strain barcode was represented by at least 64 sequencing counts in the control sample. This again allowed a range order list of gene deletion strains by magnitude of change, and pathway analysis using GeneMANIA. In conclusion: 1). yeast deletion series are a powerful tool to study pathways mediating evolution fitness for life ; 2). All milestones were met or exceeded for the flight experiment, and associated ground based static and clinorotation controls ; 3). Rank ordered lists of gene deletions have been generated by magnitude of change for suspension culture and spotted arrays ; 4). Pathway analysis of associations is ongoing ; 5). Space flight has a unique gene survival analysis footprint with several clusters of functionally interconnected genes, including clusters of genes involved in DNA repair and metabolism, transcription, chromatin remodeling, mitochondrial function, polarity, ribosome assembly, and stress ; 6). A public data website has been constructed and will be active within days to share the data.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Abstracts for Journals and Proceedings	Hammond TG, Becker J, Birdsall HH, Stodieck L, Koenig P, Johnson A, Hammond J, Allen PL. "Candida albicans virulence is reduced during spaceflight in a nematode assay." ESA/ISGP – Life in Space for Life on Earth Symposium, Aberdeen, UK, June 18-22, 2012. ESA/ISGP – Life in Space for Life on Earth Symposium, Aberdeen, UK, June 18-22, 2012. , Jun-2012
Articles in Peer-reviewed Journals	Hammond TG, Becker JL, Stodieck LS, Koenig P, Johnson A, Hammond JS, Gunter MA, Allen PL. "Suspension culture assay of bacterial virulence using Caenorhabditis elegans: optimization for space flight studies." Journal of Gravitational Physiology. In Press, as of May 2012. , May-2012